46755 – Renewables in Electricity Markets

Lecture 4: Electricity market clearing: Optimization vs. equilibrium

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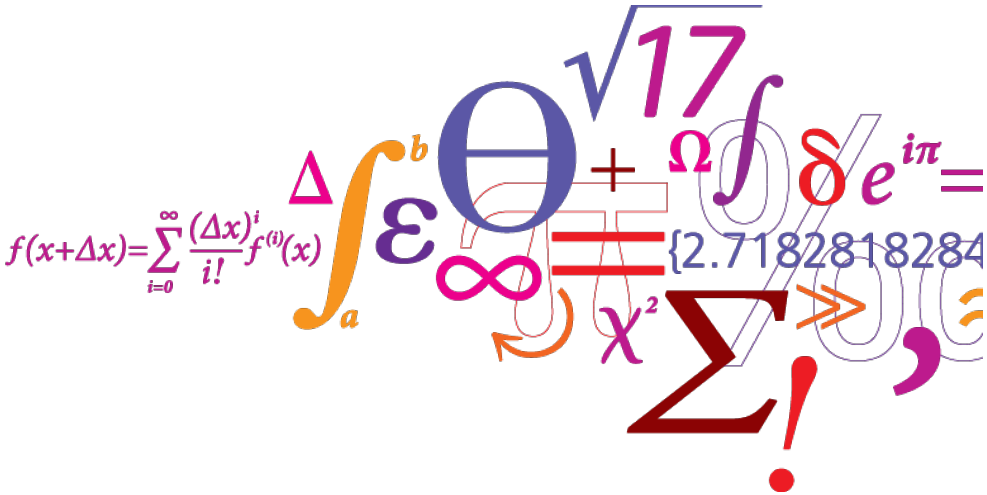

Remember from the previous lecture …

- Fundamentals of Day-Ahead Market Clearing

  Gate closure| Market time unit| Why day-ahead market: technical and economics reasons

- Single Day-Ahead Coupling (SDAC) in Europe

  Motivation| EUPHEMIA

- Network Effects on Market-Clearing Outcomes

  Different LMPs| Congestion rent | Spatial arbitrager | Budget balance | Market-clearing optimization with network constraints

- Power Flow Equations

  AC power flow equations| Non-linear and non-convex | Approximation | Linearized (DC) power flow equations | Global vs. local optimal point

- Market-Clearing Optimization Problem: Nodal vs. Zonal

  Locational marginal prices (LMPs) | Zonal prices | Net transfer capacity (NTC) | Re-dispatch | Regional coordination centers (RCCs)

- Flow-Based Market Coupling in Europe

  Critical domestic lines | Remaining available margin (RAM) | PTDFs

Market clearing: A simple example

Two generators (G1 and G2) and two price-elastic demands (D1 and D2)

Capacity: 100 MW
Offer price: €12/MWh

Capacity: 80 MW
Offer price: €20/MWh

Demand D1

Maximum load: 100 MW
Bid price: €40/MWh

Demand D2

Maximum load: 50 MW
Bid price: €35/MWh

Market outcomes:

- Productions of G1 and G2: **100 MW and 50 MW**
- Consumptions of D1 and D2: **100 MW and 50 MW**
- Market-clearing price: **€20/MWh**

Market clearing as an optimization problem

$$\underset{p^{\mathrm{G1}},p^{\mathrm{G2}},p^{\mathrm{D1}},p^{\mathrm{D2}}}{\text{Maximize}} \quad SW = [40p^{\mathrm{D1}} + 35p^{\mathrm{D2}}] - [12p^{\mathrm{G1}} + 20p^{\mathrm{G2}}] \tag{1a}$$

subject to:

$$0 \leq p^{\mathrm{D1}} \leq 100 \tag{1b}$$

$$0 \leq p^{\mathrm{D2}} \leq 50 \tag{1c}$$

$$0 \leq p^{\mathrm{G1}} \leq 100 \tag{1d}$$

$$0 \leq p^{\mathrm{G2}} \leq 80 \tag{1e}$$

$$p^{\mathrm{D1}} + p^{\mathrm{D2}} - p^{\mathrm{G1}} - p^{\mathrm{G2}} = 0 \qquad : \lambda \tag{1f}$$

Market clearing as an optimization problem

$$\underset{p^{\mathrm{G1}},p^{\mathrm{G2}},p^{\mathrm{D1}},p^{\mathrm{D2}}}{\text{Maximize}} \quad SW = \underbrace{[40p^{\mathrm{D1}} + 35p^{\mathrm{D2}}]}_{\text{Utility of demands}} - \underbrace{[12p^{\mathrm{G1}} + 20p^{\mathrm{G2}}]}_{\text{Cost of generators}} \tag{1a}$$

subject to:

$$0 \leq p^{\mathrm{D1}} \leq 100 \tag{1b}$$

$$0 \leq p^{\mathrm{D2}} \leq 50 \tag{1c}$$

Consumption limits

$$0 \leq p^{\mathrm{G1}} \leq 100 \tag{1d}$$

$$0 \leq p^{\mathrm{G2}} \leq 80 \tag{1e}$$

Generation limits

$$\underbrace{p^{\mathrm{D1}} + p^{\mathrm{D2}} - p^{\mathrm{G1}} - p^{\mathrm{G2}} = 0}_{\text{Power balance equality}} \quad : \lambda \tag{1f}$$

Dual variable: market-clearing price

Discussion

Question:

How can we ensure that all market participants (i.e., G1, G2, D1, and D2) are satisfied with the market-clearing outcome and have no incentive to deviate from it?

Discussion

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How can we ensure that all market participants (i.e., G1, G2, D1, and D2) are satisfied with the market-clearing outcome and have no incentive to deviate from it?

Let's formulate an optimization problem for each market participant!

Discussion

Question:

How can we ensure that all market participants (i.e., G1, G2, D1, and D2) are satisfied with the market-clearing outcome and have no incentive to deviate from it?

Let's formulate an optimization problem for each market participant!

Question:

What is the objective of each generator?
What is the objective of each elastic demand?

Discussion

Question:

How can we ensure that all market participants (i.e., G1, G2, D1, and D2) are satisfied with the market-clearing outcome and have no incentive to deviate from it?

Let's formulate an optimization problem for each market participant!

Question:

What is the objective of each generator? Profit maximization!

What is the objective of each elastic demand? Utility maximization!

Discussion

Question:

How can we ensure that all market participants (i.e., G1, G2, D1, and D2) are satisfied with the market-clearing outcome and have no incentive to deviate from it?

Let's formulate an optimization problem for each market participant!

Question:

What is the objective of each generator? Profit maximization!
What is the objective of each elastic demand? Utility maximization!

Question:

How to calculate a generator's profit?
How to calculate an elastic demand's utility?

Discussion

Question:

How can we ensure that all market participants (i.e., G1, G2, D1, and D2) are satisfied with the market-clearing outcome and have no incentive to deviate from it?

Question:

What is the objective of each generator? Profit maximization!

What is the objective of each elastic demand? Utility maximization!

Question:

How to calculate a generator's profit? Production level x [market price – production cost]

How to calculate an elastic demand's utility? Consumption level x [bid price – market price]

Optimization problem of each market player

Optimization problem of each market player

For generator G1:

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$

subject to:

$$0 \leq p^{\mathrm{G1}} \leq 100 \quad : \underline{\mu}^{\mathrm{G1}}, \overline{\mu}^{\mathrm{G1}}$$

Optimization problem of each market player

For generator G1:

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$

subject to:

$$0 \leq p^{\mathrm{G1}} \leq 100 \quad : \underline{\mu}^{\mathrm{G1}}, \overline{\mu}^{\mathrm{G1}}$$

For generator G2:

$$\underset{p^{\mathrm{G2}}}{\text{Maximize}} \quad p^{\mathrm{G2}}(\lambda - 20)$$

subject to:

$$0 \leq p^{\mathrm{G2}} \leq 80 \quad : \underline{\mu}^{\mathrm{G2}}, \overline{\mu}^{\mathrm{G2}}$$

Optimization problem of each market player

For generator G1:

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$

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$$\underset{p^{\mathrm{G2}}}{\text{Maximize}} \quad p^{\mathrm{G2}}(\lambda - 20)$$

subject to:

$$0 \leq p^{\mathrm{G2}} \leq 80 \quad : \underline{\mu}^{\mathrm{G2}}, \overline{\mu}^{\mathrm{G2}}$$

For demand D1:

$$\underset{p^{\mathrm{D1}}}{\text{Maximize}} \quad p^{\mathrm{D1}}(40 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D1}} \leq 100 \quad : \underline{\mu}^{\mathrm{D1}}, \overline{\mu}^{\mathrm{D1}}$$

Optimization problem of each market player

For generator G1:

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$

subject to:

$$0 \leq p^{\mathrm{G1}} \leq 100 \quad : \underline{\mu}^{\mathrm{G1}}, \overline{\mu}^{\mathrm{G1}}$$

For generator G2:

$$\underset{p^{\mathrm{G2}}}{\text{Maximize}} \quad p^{\mathrm{G2}}(\lambda - 20)$$

subject to:

$$0 \leq p^{\mathrm{G2}} \leq 80 \quad : \underline{\mu}^{\mathrm{G2}}, \overline{\mu}^{\mathrm{G2}}$$

For demand D1:

$$\underset{p^{\mathrm{D1}}}{\text{Maximize}} \quad p^{\mathrm{D1}}(40 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D1}} \leq 100 \quad : \underline{\mu}^{\mathrm{D1}}, \overline{\mu}^{\mathrm{D1}}$$

For demand D2:

$$\underset{p^{\mathrm{D2}}}{\text{Maximize}} \quad p^{\mathrm{D2}}(35 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D2}} \leq 50 \quad : \underline{\mu}^{\mathrm{D2}}, \overline{\mu}^{\mathrm{D2}}$$

Optimization problem of each market player

For generator G1:

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$

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The market price λ is treated as a given parameter in each optimization problem!

Optimization problem of each market player

For generator G1:

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$

subject to:

$$0 \leq p^{\mathrm{G1}} \leq 100 \quad : \underline{\mu}^{\mathrm{G1}}, \overline{\mu}^{\mathrm{G1}}$$

For generator G2:

$$\underset{p^{\mathrm{G2}}}{\text{Maximize}} \quad p^{\mathrm{G2}}(\lambda - 20)$$

subject to:

$$0 \leq p^{\mathrm{G2}} \leq 80 \quad : \underline{\mu}^{\mathrm{G2}}, \overline{\mu}^{\mathrm{G2}}$$

For demand D1:

$$\underset{p^{\mathrm{D1}}}{\text{Maximize}} \quad p^{\mathrm{D1}}(40 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D1}} \leq 100 \quad : \underline{\mu}^{\mathrm{D1}}, \overline{\mu}^{\mathrm{D1}}$$

For demand D2:

$$\underset{p^{\mathrm{D2}}}{\text{Maximize}} \quad p^{\mathrm{D2}}(35 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D2}} \leq 50 \quad : \underline{\mu}^{\mathrm{D2}}, \overline{\mu}^{\mathrm{D2}}$$

Question:
How do market players contribute to market price formation? Is an additional condition needed?

Optimization problem of each market player

For generator G1:

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$

subject to:

$$0 \leq p^{\mathrm{G1}} \leq 100 \quad : \underline{\mu}^{\mathrm{G1}}, \overline{\mu}^{\mathrm{G1}}$$

For generator G2:

$$\underset{p^{\mathrm{G2}}}{\text{Maximize}} \quad p^{\mathrm{G2}}(\lambda - 20)$$

subject to:

$$0 \leq p^{\mathrm{G2}} \leq 80 \quad : \underline{\mu}^{\mathrm{G2}}, \overline{\mu}^{\mathrm{G2}}$$

For demand D1:

$$\underset{p^{\mathrm{D1}}}{\text{Maximize}} \quad p^{\mathrm{D1}}(40 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D1}} \leq 100 \quad : \underline{\mu}^{\mathrm{D1}}, \overline{\mu}^{\mathrm{D1}}$$

For demand D2:

$$\underset{p^{\mathrm{D2}}}{\text{Maximize}} \quad p^{\mathrm{D2}}(35 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D2}} \leq 50 \quad : \underline{\mu}^{\mathrm{D2}}, \overline{\mu}^{\mathrm{D2}}$$

Question:
How do market players contribute to market price formation? Is an additional condition needed? Yes!

Optimization problem of each market player

For generator G1:

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$

subject to:

$$0 \leq p^{\mathrm{G1}} \leq 100 \quad : \underline{\mu}^{\mathrm{G1}}, \overline{\mu}^{\mathrm{G1}}$$

For generator G2:

$$\underset{p^{\mathrm{G2}}}{\text{Maximize}} \quad p^{\mathrm{G2}}(\lambda - 20)$$

subject to:

$$0 \leq p^{\mathrm{G2}} \leq 80 \quad : \underline{\mu}^{\mathrm{G2}}, \overline{\mu}^{\mathrm{G2}}$$

For demand D1:

$$\underset{p^{\mathrm{D1}}}{\text{Maximize}} \quad p^{\mathrm{D1}}(40 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D1}} \leq 100 \quad : \underline{\mu}^{\mathrm{D1}}, \overline{\mu}^{\mathrm{D1}}$$

For demand D2:

$$\underset{p^{\mathrm{D2}}}{\text{Maximize}} \quad p^{\mathrm{D2}}(35 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D2}} \leq 50 \quad : \underline{\mu}^{\mathrm{D2}}, \overline{\mu}^{\mathrm{D2}}$$

Power balance equality:

$$p^{\mathrm{D1}} + p^{\mathrm{D2}} - p^{\mathrm{G1}} - p^{\mathrm{G2}} = 0 \quad : \lambda$$

Optimization problem of each market player

For generator G1:

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$

subject to:

$$0 \leq p^{\mathrm{G1}} \leq 100 \quad : \underline{\mu}^{\mathrm{G1}}, \overline{\mu}^{\mathrm{G1}}$$

For generator G2:

$$\underset{p^{\mathrm{G2}}}{\text{Maximize}} \quad p^{\mathrm{G2}}(\lambda - 20)$$

subject to:

$$0 \leq p^{\mathrm{G2}} \leq 80 \quad : \underline{\mu}^{\mathrm{G2}}, \overline{\mu}^{\mathrm{G2}}$$

For demand D1:

$$\underset{p^{\mathrm{D1}}}{\text{Maximize}} \quad p^{\mathrm{D1}}(40 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D1}} \leq 100 \quad : \underline{\mu}^{\mathrm{D1}}, \overline{\mu}^{\mathrm{D1}}$$

For demand D2:

$$\underset{p^{\mathrm{D2}}}{\text{Maximize}} \quad p^{\mathrm{D2}}(35 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D2}} \leq 50 \quad : \underline{\mu}^{\mathrm{D2}}, \overline{\mu}^{\mathrm{D2}}$$

Power balance equality:

$$p^{\mathrm{D1}} + p^{\mathrm{D2}} - p^{\mathrm{G1}} - p^{\mathrm{G2}} = 0 \quad : \lambda$$

Question: From a mathematical perspective, is the power balance equality above **equivalent** to the following **unconstrained** optimization problem?

$$\underset{\lambda}{\text{Maximize}} \quad \lambda \left(p^{\mathrm{D1}} + p^{\mathrm{D2}} - p^{\mathrm{G1}} - p^{\mathrm{G2}}\right)$$

Optimization problem of each market player

For generator G1:

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$

subject to:

$$0 \leq p^{\mathrm{G1}} \leq 100 \quad : \underline{\mu}^{\mathrm{G1}}, \overline{\mu}^{\mathrm{G1}}$$

For generator G2:

$$\underset{p^{\mathrm{G2}}}{\text{Maximize}} \quad p^{\mathrm{G2}}(\lambda - 20)$$

subject to:

$$0 \leq p^{\mathrm{G2}} \leq 80 \quad : \underline{\mu}^{\mathrm{G2}}, \overline{\mu}^{\mathrm{G2}}$$

For demand D1:

$$\underset{p^{\mathrm{D1}}}{\text{Maximize}} \quad p^{\mathrm{D1}}(40 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D1}} \leq 100 \quad : \underline{\mu}^{\mathrm{D1}}, \overline{\mu}^{\mathrm{D1}}$$

For demand D2:

$$\underset{p^{\mathrm{D2}}}{\text{Maximize}} \quad p^{\mathrm{D2}}(35 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D2}} \leq 50 \quad : \underline{\mu}^{\mathrm{D2}}, \overline{\mu}^{\mathrm{D2}}$$

Power balance equality:

$$p^{\mathrm{D1}} + p^{\mathrm{D2}} - p^{\mathrm{G1}} - p^{\mathrm{G2}} = 0 \quad : \lambda$$

Yes, derive KKTs!

$$\underset{\lambda}{\text{Maximize}} \quad \lambda \left(p^{\mathrm{D1}} + p^{\mathrm{D2}} - p^{\mathrm{G1}} - p^{\mathrm{G2}}\right)$$

Optimization problem of each market player

For generator G1:

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$

subject to:

$$0 \leq p^{\mathrm{G1}} \leq 100 \quad : \underline{\mu}^{\mathrm{G1}}, \overline{\mu}^{\mathrm{G1}}$$

For generator G2:

$$\underset{p^{\mathrm{G2}}}{\text{Maximize}} \quad p^{\mathrm{G2}}(\lambda - 20)$$

subject to:

$$0 \leq p^{\mathrm{G2}} \leq 80 \quad : \underline{\mu}^{\mathrm{G2}}, \overline{\mu}^{\mathrm{G2}}$$

For demand D1:

$$\underset{p^{\mathrm{D1}}}{\text{Maximize}} \quad p^{\mathrm{D1}}(40 - \lambda)$$

subject to:

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For demand D2:

$$\underset{p^{\mathrm{D2}}}{\text{Maximize}} \quad p^{\mathrm{D2}}(35 - \lambda)$$

subject to:

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Power balance equality:

$$p^{\mathrm{D1}} + p^{\mathrm{D2}} - p^{\mathrm{G1}} - p^{\mathrm{G2}} = 0 \quad : \lambda$$

Assume a fictitious player, the so-called "**price-setter**," solves this unconstrained optimization problem to determine the market-clearing price.

$$\underset{\lambda}{\text{Maximize}} \quad \lambda \left(p^{\mathrm{D1}} + p^{\mathrm{D2}} - p^{\mathrm{G1}} - p^{\mathrm{G2}}\right)$$

Optimization problem of each market player

For generator G1:

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$

subject to:

$$0 \leq p^{\mathrm{G1}} \leq 100 \quad : \underline{\mu}^{\mathrm{G1}}, \overline{\mu}^{\mathrm{G1}}$$

For generator G2:

$$\underset{p^{\mathrm{G2}}}{\text{Maximize}} \quad p^{\mathrm{G2}}(\lambda - 20)$$

subject to:

$$0 \leq p^{\mathrm{G2}} \leq 80 \quad : \underline{\mu}^{\mathrm{G2}}, \overline{\mu}^{\mathrm{G2}}$$

For demand D1:

$$\underset{p^{\mathrm{D1}}}{\text{Maximize}} \quad p^{\mathrm{D1}}(40 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D1}} \leq 100 \quad : \underline{\mu}^{\mathrm{D1}}, \overline{\mu}^{\mathrm{D1}}$$

For demand D2:

$$\underset{p^{\mathrm{D2}}}{\text{Maximize}} \quad p^{\mathrm{D2}}(35 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D2}} \leq 50 \quad : \underline{\mu}^{\mathrm{D2}}, \overline{\mu}^{\mathrm{D2}}$$

Price-setter:

$$\underset{\lambda}{\text{Maximize}} \quad \lambda \left(p^{\mathrm{D1}} + p^{\mathrm{D2}} - p^{\mathrm{G1}} - p^{\mathrm{G2}}\right)$$

Optimization problem of each market player

For generator G1:

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$

subject to:

$$0 \leq p^{\mathrm{G1}} \leq 100 \quad : \underline{\mu}^{\mathrm{G1}}, \overline{\mu}^{\mathrm{G1}}$$

For generator G2:

$$\underset{p^{\mathrm{G2}}}{\text{Maximize}} \quad p^{\mathrm{G2}}(\lambda - 20)$$

subject to:

$$0 \leq p^{\mathrm{G2}} \leq 80 \quad : \underline{\mu}^{\mathrm{G2}}, \overline{\mu}^{\mathrm{G2}}$$

For demand D1:

$$\underset{p^{\mathrm{D1}}}{\text{Maximize}} \quad p^{\mathrm{D1}}(40 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D1}} \leq 100 \quad : \underline{\mu}^{\mathrm{D1}}, \overline{\mu}^{\mathrm{D1}}$$

For demand D2:

$$\underset{p^{\mathrm{D2}}}{\text{Maximize}} \quad p^{\mathrm{D2}}(35 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D2}} \leq 50 \quad : \underline{\mu}^{\mathrm{D2}}, \overline{\mu}^{\mathrm{D2}}$$

Price-setter:

$$\underset{\lambda}{\text{Maximize}} \quad \lambda \left(p^{\mathrm{D1}} + p^{\mathrm{D2}} - p^{\mathrm{G1}} - p^{\mathrm{G2}}\right)$$

Question: Can we solve optimization problems above separately?

Optimization problem of each market player

For generator G1:

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$

subject to:

$$0 \leq p^{\mathrm{G1}} \leq 100 \quad : \underline{\mu}^{\mathrm{G1}}, \overline{\mu}^{\mathrm{G1}}$$

For generator G2:

$$\underset{p^{\mathrm{G2}}}{\text{Maximize}} \quad p^{\mathrm{G2}}(\lambda - 20)$$

subject to:

$$0 \leq p^{\mathrm{G2}} \leq 80 \quad : \underline{\mu}^{\mathrm{G2}}, \overline{\mu}^{\mathrm{G2}}$$

For demand D1:

$$\underset{p^{\mathrm{D1}}}{\text{Maximize}} \quad p^{\mathrm{D1}}(40 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D1}} \leq 100 \quad : \underline{\mu}^{\mathrm{D1}}, \overline{\mu}^{\mathrm{D1}}$$

For demand D2:

$$\underset{p^{\mathrm{D2}}}{\text{Maximize}} \quad p^{\mathrm{D2}}(35 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D2}} \leq 50 \quad : \underline{\mu}^{\mathrm{D2}}, \overline{\mu}^{\mathrm{D2}}$$

Price-setter:

$$\underset{\lambda}{\text{Maximize}} \quad \lambda \left(p^{\mathrm{D1}} + p^{\mathrm{D2}} - p^{\mathrm{G1}} - p^{\mathrm{G2}}\right)$$

Question: Can we solve optimization problems above separately? No! Why?

Optimization problem of each market player

For generator G1:

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$

subject to:

$$0 \leq p^{\mathrm{G1}} \leq 100 \quad : \underline{\mu}^{\mathrm{G1}}, \overline{\mu}^{\mathrm{G1}}$$

For generator G2:

$$\underset{p^{\mathrm{G2}}}{\text{Maximize}} \quad p^{\mathrm{G2}}(\lambda - 20)$$

subject to:

$$0 \leq p^{\mathrm{G2}} \leq 80 \quad : \underline{\mu}^{\mathrm{G2}}, \overline{\mu}^{\mathrm{G2}}$$

For demand D1:

$$\underset{p^{\mathrm{D1}}}{\text{Maximize}} \quad p^{\mathrm{D1}}(40 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D1}} \leq 100 \quad : \underline{\mu}^{\mathrm{D1}}, \overline{\mu}^{\mathrm{D1}}$$

For demand D2:

$$\underset{p^{\mathrm{D2}}}{\text{Maximize}} \quad p^{\mathrm{D2}}(35 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D2}} \leq 50 \quad : \underline{\mu}^{\mathrm{D2}}, \overline{\mu}^{\mathrm{D2}}$$

Price-setter:

$$\underset{\lambda}{\text{Maximize}} \quad \lambda \left(p^{\mathrm{D1}} + p^{\mathrm{D2}} - p^{\mathrm{G1}} - p^{\mathrm{G2}}\right)$$

- The market-clearing price is a <u>variable</u> for the price-setter, but a <u>parameter</u> for G1, G2, D1, and D2.
- Productions/consumptions are <u>variables</u> for G1, G2, D1 and D2, but <u>parameters</u> for the price-setter.

Optimization problem of each market player

For generator G1:

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$

subject to:

$$0 \leq p^{\mathrm{G1}} \leq 100 \quad : \underline{\mu}^{\mathrm{G1}}, \overline{\mu}^{\mathrm{G1}}$$

For generator G2:

$$\underset{p^{\mathrm{G2}}}{\text{Maximize}} \quad p^{\mathrm{G2}}(\lambda - 20)$$

subject to:

$$0 \leq p^{\mathrm{G2}} \leq 80 \quad : \underline{\mu}^{\mathrm{G2}}, \overline{\mu}^{\mathrm{G2}}$$

For demand D1:

$$\underset{p^{\mathrm{D1}}}{\text{Maximize}} \quad p^{\mathrm{D1}}(40 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D1}} \leq 100 \quad : \underline{\mu}^{\mathrm{D1}}, \overline{\mu}^{\mathrm{D1}}$$

For demand D2:

$$\underset{p^{\mathrm{D2}}}{\text{Maximize}} \quad p^{\mathrm{D2}}(35 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D2}} \leq 50 \quad : \underline{\mu}^{\mathrm{D2}}, \overline{\mu}^{\mathrm{D2}}$$

Price-setter:

$$\underset{\lambda}{\text{Maximize}} \quad \lambda \left(p^{\mathrm{D1}} + p^{\mathrm{D2}} - p^{\mathrm{G1}} - p^{\mathrm{G2}}\right)$$

All five optimization problems above are <u>interconnected</u> and should be solved together!

Optimization problem of each market player

For generator G1:

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$

subject to:

$$0 \leq p^{\mathrm{G1}} \leq 100 \quad : \underline{\mu}^{\mathrm{G1}}, \overline{\mu}^{\mathrm{G1}}$$

For generator G2:

$$\underset{p^{\mathrm{G2}}}{\text{Maximize}} \quad p^{\mathrm{G2}}(\lambda - 20)$$

subject to:

$$0 \leq p^{\mathrm{G2}} \leq 80 \quad : \underline{\mu}^{\mathrm{G2}}, \overline{\mu}^{\mathrm{G2}}$$

For demand D1:

$$\underset{p^{\mathrm{D1}}}{\text{Maximize}} \quad p^{\mathrm{D1}}(40 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D1}} \leq 100 \quad : \underline{\mu}^{\mathrm{D1}}, \overline{\mu}^{\mathrm{D1}}$$

For demand D2:

$$\underset{p^{\mathrm{D2}}}{\text{Maximize}} \quad p^{\mathrm{D2}}(35 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D2}} \leq 50 \quad : \underline{\mu}^{\mathrm{D2}}, \overline{\mu}^{\mathrm{D2}}$$

Price-setter:

$$\underset{\lambda}{\text{Maximize}} \quad \lambda \left(p^{\mathrm{D1}} + p^{\mathrm{D2}} - p^{\mathrm{G1}} - p^{\mathrm{G2}}\right)$$

This is a game-theoretic problem.

Optimization problem of each market player

For generator G1:

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$

subject to:

$$0 \leq p^{\mathrm{G1}} \leq 100 \quad : \underline{\mu}^{\mathrm{G1}}, \overline{\mu}^{\mathrm{G1}}$$

For generator G2:

$$\underset{p^{\mathrm{G2}}}{\text{Maximize}} \quad p^{\mathrm{G2}}(\lambda - 20)$$

subject to:

$$0 \leq p^{\mathrm{G2}} \leq 80 \quad : \underline{\mu}^{\mathrm{G2}}, \overline{\mu}^{\mathrm{G2}}$$

For demand D1:

$$\underset{p^{\mathrm{D1}}}{\text{Maximize}} \quad p^{\mathrm{D1}}(40 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D1}} \leq 100 \quad : \underline{\mu}^{\mathrm{D1}}, \overline{\mu}^{\mathrm{D1}}$$

For demand D2:

$$\underset{p^{\mathrm{D2}}}{\text{Maximize}} \quad p^{\mathrm{D2}}(35 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D2}} \leq 50 \quad : \underline{\mu}^{\mathrm{D2}}, \overline{\mu}^{\mathrm{D2}}$$

Price-setter:

$$\underset{\lambda}{\text{Maximize}} \quad \lambda \left(p^{\mathrm{D1}} + p^{\mathrm{D2}} - p^{\mathrm{G1}} - p^{\mathrm{G2}}\right)$$

This is a game-theoretic problem.

- Do the players influence each other's objective functions?

Optimization problem of each market player

For generator G1:

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$

subject to:

$$0 \leq p^{\mathrm{G1}} \leq 100 \quad : \underline{\mu}^{\mathrm{G1}}, \overline{\mu}^{\mathrm{G1}}$$

For generator G2:

$$\underset{p^{\mathrm{G2}}}{\text{Maximize}} \quad p^{\mathrm{G2}}(\lambda - 20)$$

subject to:

$$0 \leq p^{\mathrm{G2}} \leq 80 \quad : \underline{\mu}^{\mathrm{G2}}, \overline{\mu}^{\mathrm{G2}}$$

For demand D1:

$$\underset{p^{\mathrm{D1}}}{\text{Maximize}} \quad p^{\mathrm{D1}}(40 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D1}} \leq 100 \quad : \underline{\mu}^{\mathrm{D1}}, \overline{\mu}^{\mathrm{D1}}$$

For demand D2:

$$\underset{p^{\mathrm{D2}}}{\text{Maximize}} \quad p^{\mathrm{D2}}(35 - \lambda)$$

subject to:

$$0 \leq p^{\mathrm{D2}} \leq 50 \quad : \underline{\mu}^{\mathrm{D2}}, \overline{\mu}^{\mathrm{D2}}$$

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$$\underset{\lambda}{\text{Maximize}} \quad \lambda \left(p^{\mathrm{D1}} + p^{\mathrm{D2}} - p^{\mathrm{G1}} - p^{\mathrm{G2}}\right)$$

This is a game-theoretic problem.

- Do the players influence each other's objective functions? Yes

Optimization problem of each market player

For generator G1:

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$

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Price-setter:

$$\underset{\lambda}{\text{Maximize}} \quad \lambda \left(p^{\mathrm{D1}} + p^{\mathrm{D2}} - p^{\mathrm{G1}} - p^{\mathrm{G2}}\right)$$

This is a game-theoretic problem.

- Do the players influence each other's objective functions? Yes
- Do they also affect each other's feasible spaces (constraints)?

Optimization problem of each market player

For generator G1:

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$

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This is a game-theoretic problem.

- Do the players influence each other's objective functions? Yes
- Do they also affect each other's feasible spaces (constraints)? No

Optimization problem of each market player

For generator G1:

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$

subject to:

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For generator G2:

$$\underset{p^{\mathrm{G2}}}{\text{Maximize}} \quad p^{\mathrm{G2}}(\lambda - 20)$$

subject to:

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subject to:

$$0 \leq p^{\mathrm{D1}} \leq 100 \quad : \underline{\mu}^{\mathrm{D1}}, \overline{\mu}^{\mathrm{D1}}$$

For demand D2:

$$\underset{p^{\mathrm{D2}}}{\text{Maximize}} \quad p^{\mathrm{D2}}(3$$

subject to:

$$0 \leq p^{\mathrm{D2}} \leq 50 \quad :$$

Price-setter:

$$\underset{\lambda}{\text{Maximize}} \quad \lambda \left(p^{\mathrm{D1}} + p^{\mathrm{D2}} - p^{\mathrm{G1}} - p^{\mathrm{G2}}\right)$$

This is a game-theoretic problem.

- Do the players influence each other's objective functions? Yes
- Do they also affect each other's feasible spaces (constraints)? No

This is a **"Nash equilibrium" (NE)** game, and we can explore the existence and uniqueness of the equilibrium solution.

Otherwise, if impacting each other's feasible spaces, it would be a **Generalized Nash Equilibrium (GNE)** game, whose properties are generally difficult to investigate.

Optimization problem of each market player

For generator G1:

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$

subject to:

$$0 \leq p^{\mathrm{G1}} \leq 100 \quad : \underline{\mu}^{\mathrm{G1}}, \overline{\mu}^{\mathrm{G1}}$$

For generator G2:

$$\underset{p^{\mathrm{G2}}}{\text{Maximize}} \quad p^{\mathrm{G2}}(\lambda - 20)$$

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For demand D2:

$$\underset{p^{\mathrm{D2}}}{\text{Maximize}} \quad p^{\mathrm{D2}}(35 - \lambda)$$

subject to:

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Price-setter:

$$\underset{\lambda}{\text{Maximize}} \quad \lambda \left(p^{\mathrm{D1}} + p^{\mathrm{D2}} - p^{\mathrm{G1}} - p^{\mathrm{G2}}\right)$$

This is a game-theoretic problem.

This specific problem is also referred to as the "competitive equilibrium" problem!

Seminal works on competitive equilibrium

L. V. Kantorovich, "Mathematical methods of organizing and planning production," *Management Science*, vol. 6, no. 4, pp. 366–422, 1960.

MATHEMATICAL METHODS OF ORGANIZING AND PLANNING PRODUCTION*†

L. V. KANTOROVICH

Leningrad State University

1939

Contents

Leonid Kantorovich
Nobel Prize in Economic Sciences (1975)

P. A. Samuelson, "Spatial price equilibrium and linear programming," *American Economic Review*, vol. 42, no. 3, pp. 283–303, 1952.

SPATIAL PRICE EQUILIBRIUM AND LINEAR PROGRAMMING

By PAUL A. SAMUELSON*

I.—*Introduction*

Increasingly, modern economic theorists are going beyond the formulation of equilibrium in terms of such marginal equalities as marginal revenue equal to marginal costs or wage rate equal to marginal value product. Instead they are reverting to an earlier and more fundamental aspect of a maximum position: namely, that from the top of a hill, whether or not it is locally flat, all movements are downward.

Paul Samuelson
Nobel Prize in Economic Sciences (1970)

K. J. Arrow and G. Debreu, "Existence of an equilibrium for a competitive economy," *Econometrica*, vol. 22, no. 3, pp. 265–290,1954.

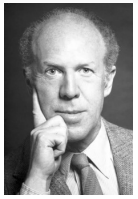
ECONOMETRICA

VOLUME 22 July, 1954 NUMBER 3

EXISTENCE OF AN EQUILIBRIUM FOR A COMPETITIVE ECONOMY

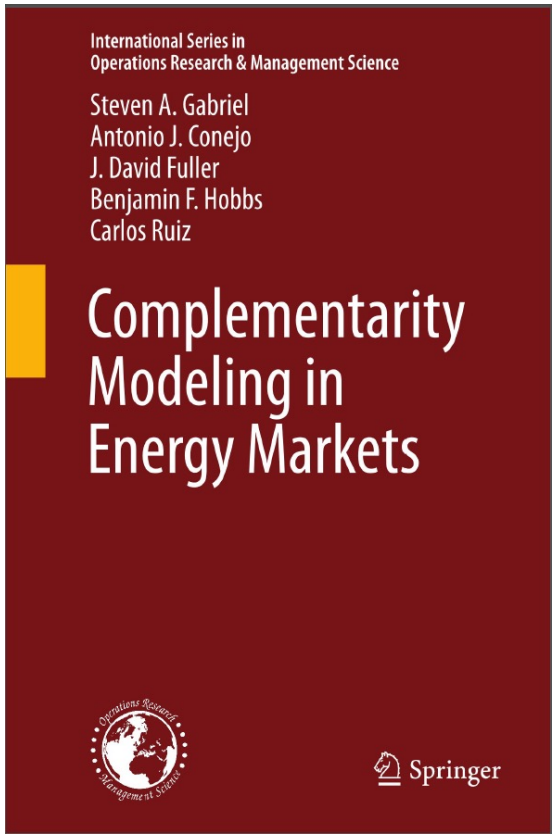
BY KENNETH J. ARROW AND GERARD DEBREU[1]

A. Wald has presented a model of production and a model of exchange and proofs of the existence of an equilibrium for each of them. Here proofs of the existence of an equilibrium are given for an *integrated* model of production, exchange and consumption. In addition the assumptions made on the technologies of producers and the tastes of consumers are significantly weaker than Wald's. Finally a simplification of the structure of the proofs has been made possible through use of the concept of an abstract economy, a generalization of that of a game.



Kenneth Arrow & Gérard Debreu
Nobel Prize in Economic Sciences (1972, 1983)

Relevant book and course!



- Prof. Steven Gabriel's yearly short course at NTNU, "**Introduction Course in Complementarity Models and Equilibrium**": https://www.ntnu.edu/studies/courses/I%C3%988806#tab=omEmnet

Market clearing as an equilibrium problem

- If a solution to our equilibrium problem exists, it will be a "Nash equilibrium point", i.e.,

John Nash
Nobel Prize in Economic Sciences (1994)

No market participant can increase its profit by deviating unilaterally from the equilibrium solution!

Market clearing as an equilibrium problem

- If a solution to our equilibrium problem exists, it will be a "Nash equilibrium point", i.e.,

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Recall the first question:
How can we ensure that all market participants (i.e., G1, G2, D1, and D2) are satisfied with the outcomes of the market-clearing optimization problem and have no incentive to deviate from it?

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Recall the first question:
How can we ensure that all market participants (i.e., G1, G2, D1, and D2) are satisfied with the outcomes of the market-clearing optimization problem and have no incentive to deviate from it?

And new questions:
- Which problem should we solve to clear the market: Optimization or equilibrium?
- How can we solve an equilibrium problem?

Market clearing: Optimization vs. equilibrium

Equilibrium

For each generator:

Maximize profit

subject to production limits

For each demand:

Maximize utility

subject to consumption limits

Price-setter's problem

Optimization

Maximize market's social welfare

subject to:

- Production limits of generators
- Consumption limits of demands
- Power balance

How to solve the equilibrium problem?

Equilibrium

For each generator:

Maximize profit

subject to production limits

For each demand:

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Price-setter's problem

How to solve the equilibrium problem?

Equilibrium

For each generator:

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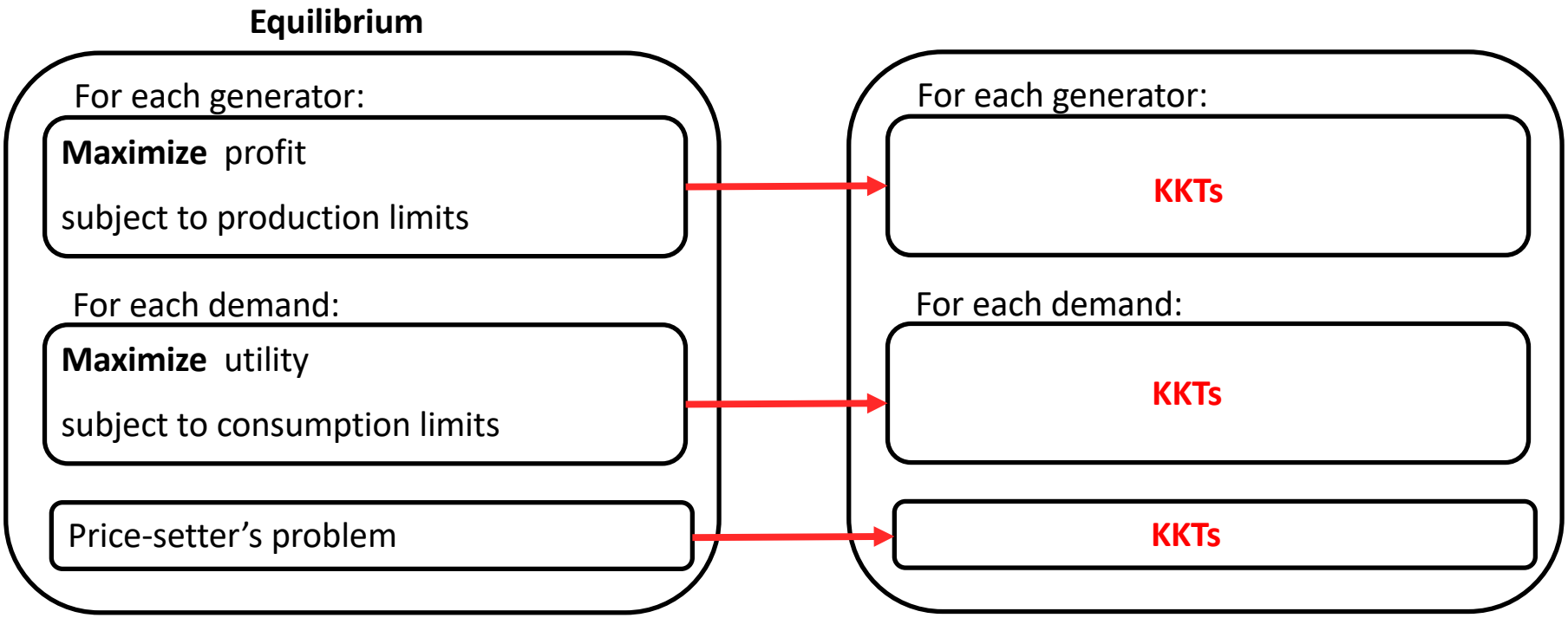
Price-setter's problem

Replace each optimization problem within the equilibrium problem with its equivalent Karush-Kuhn-Tucker (KKT) optimality conditions. Remember that these conditions consist of a set of equality and inequality constraints without any objective function!

How to solve the equilibrium problem?



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How to solve the equilibrium problem?

Equilibrium

For each generator:

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Price-setter's problem

For each generator:

KKTs

For each demand:

KKTs

KKTs

Mixed complementarity problem (MCP)

Market clearing: Optimization vs. equilibrium

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For each generator:

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Price-setter's problem

Optimization

Maximize market's social welfare

subject to:

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- Consumption limits of demands
- Power balance

Let's check again the equilibrium and optimization problems above!

Market clearing: Optimization vs. equilibrium

Equilibrium

For each generator:

Maximize profit

subject to production limits

For each demand:

Maximize utility

subject to consumption limits

Price-setter's problem

MCP

Optimization

Maximize market's social welfare

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KKTs

Market clearing: Optimization vs. equilibrium

Equilibrium

For each generator:

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subject to consumption limits

Price-setter's problem

Optimization

Maximize market's social welfare

subject to:

- Production limits of generators
- Consumption limits of demands
- Power balance

MCP

KKTs

Do the MCP (obtained from equilibrium) and the KKT conditions (obtained from optimization) include **identical** conditions?

If they do, then the equilibrium and optimization problems mentioned above are considered "**equivalent**", meaning that any solution to the equilibrium problem is also a solution to the optimization problem, and vice versa.

Market clearing as an equilibrium problem

$$\underset{p^{\mathrm{G1}}}{\text{Maximize}} \quad p^{\mathrm{G1}}(\lambda - 12)$$
subject to:
$$0 \leq p^{\mathrm{G1}} \leq 100 \quad : \underline{\mu}^{\mathrm{G1}}, \overline{\mu}^{\mathrm{G1}}$$

$$\underset{p^{\mathrm{G2}}}{\text{Maximize}} \quad p^{\mathrm{G2}}(\lambda - 20)$$
subject to:
$$0 \leq p^{\mathrm{G2}} \leq 80 \quad : \underline{\mu}^{\mathrm{G2}}, \overline{\mu}^{\mathrm{G2}}$$

$$\underset{p^{\mathrm{D1}}}{\text{Maximize}} \quad p^{\mathrm{D1}}(40 - \lambda)$$
subject to:
$$0 \leq p^{\mathrm{D1}} \leq 100 \quad : \underline{\mu}^{\mathrm{D1}}, \overline{\mu}^{\mathrm{D1}}$$

$$\underset{p^{\mathrm{D2}}}{\text{Maximize}} \quad p^{\mathrm{D2}}(35 - \lambda)$$
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$$0 \leq p^{\mathrm{D2}} \leq 50 \quad : \underline{\mu}^{\mathrm{D2}}, \overline{\mu}^{\mathrm{D2}}$$

$$\underset{\lambda}{\text{Maximize}} \quad \lambda \left(p^{\mathrm{D1}} + p^{\mathrm{D2}} - p^{\mathrm{G1}} - p^{\mathrm{G2}}\right)$$

Market clearing as an equilibrium problem

$$\underset{p^{G1}}{\text{Maximize}} \quad p^{G1}(\lambda - 12)$$
subject to:
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KKTs →

$$12 - \lambda - \underline{\mu}^{G1} + \overline{\mu}^{G1} = 0$$
$$0 \le p^{G1} \perp \underline{\mu}^{G1} \ge 0$$
$$0 \le (100 - p^{G1}) \perp \overline{\mu}^{G1} \ge 0$$

$$\underset{p^{G2}}{\text{Maximize}} \quad p^{G2}(\lambda - 20)$$
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KKTs →

$$20 - \lambda - \underline{\mu}^{G2} + \overline{\mu}^{G2} = 0$$
$$0 \le p^{G2} \perp \underline{\mu}^{G2} \ge 0$$
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$$\underset{\lambda}{\text{Maximize}} \quad \lambda \left(p^{D1} + p^{D2} - p^{G1} - p^{G2}\right)$$

KKTs →

$$p^{D1} + p^{D2} - p^{G1} - p^{G2} = 0$$

Market clearing as an equilibrium problem

$$\underset{p^{G1}}{\text{Maximize}} \quad p^{G1}(\lambda - 12)$$
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$$\underset{\lambda}{\text{Maximize}} \quad \lambda\left(p^{D1} + p^{D2} - p^{G1} - p^{G2}\right)$$

KKTs

$$p^{D1} + p^{D2} - p^{G1} - p^{G2} = 0$$

Mixed complementarity problem (MCP)

Market clearing as an equilibrium problem

$$12 - \lambda - \underline{\mu}^{\mathrm{G1}} + \overline{\mu}^{\mathrm{G1}} = 0$$
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$$0 \leq (50 - p^{\mathrm{D2}}) \perp \overline{\mu}^{\mathrm{D2}} \geq 0$$

$$p^{\mathrm{D1}} + p^{\mathrm{D2}} - p^{\mathrm{G1}} - p^{\mathrm{G2}} = 0$$

Mixed complementarity problem (MCP)

Market clearing as an equilibrium problem

Question:

Is this MCP identical to the KKTs of the market-clearing <u>optimization</u> problem? Let's verify!

$$12 - \lambda - \underline{\mu}^{\mathrm{G1}} + \overline{\mu}^{\mathrm{G1}} = 0$$
$$0 \leq p^{\mathrm{G1}} \perp \underline{\mu}^{\mathrm{G1}} \geq 0$$
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Mixed complementarity problem (MCP)

Market clearing as an equilibrium problem

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Is this MCP identical to the KKTs of the market-clearing <u>optimization</u> problem? Let's verify!

Answer: ?

Optimization

$$\underset{p^{\mathrm{G1}},p^{\mathrm{G2}},p^{\mathrm{D1}},p^{\mathrm{D2}}}{\text{Maximize}} \quad SW = [40p^{\mathrm{D1}} + 35p^{\mathrm{D2}}] - [12p^{\mathrm{G1}} + 20p^{\mathrm{G2}}] \tag{1a}$$

subject to:

$$0 \le p^{\mathrm{D1}} \le 100 \quad : \underline{\mu}^{\mathrm{D1}}, \overline{\mu}^{\mathrm{D1}} \tag{1b}$$

$$0 \le p^{\mathrm{D2}} \le 50 \quad : \underline{\mu}^{\mathrm{D2}}, \overline{\mu}^{\mathrm{D2}} \tag{1c}$$

$$0 \le p^{\mathrm{G1}} \le 100 \quad : \underline{\mu}^{\mathrm{G1}}, \overline{\mu}^{\mathrm{G1}} \tag{1d}$$

$$0 \le p^{\mathrm{G2}} \le 80 \quad : \underline{\mu}^{\mathrm{G2}}, \overline{\mu}^{\mathrm{G2}} \tag{1e}$$

$$p^{\mathrm{D1}} + p^{\mathrm{D2}} - p^{\mathrm{G1}} - p^{\mathrm{G2}} = 0 \quad : \lambda \tag{1f}$$

$$12 - \lambda - \underline{\mu}^{\mathrm{G1}} + \overline{\mu}^{\mathrm{G1}} = 0$$
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KKTs

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$$0 \le p^{D1} \le 100 \quad : \underline{\mu}^{D1}, \overline{\mu}^{D1} \tag{1b}$$

$$0 \le p^{D2} \le 50 \quad : \underline{\mu}^{D2}, \overline{\mu}^{D2} \tag{1c}$$

$$0 \le p^{G1} \le 100 \quad : \underline{\mu}^{G1}, \overline{\mu}^{G1} \tag{1d}$$

$$0 \le p^{G2} \le 80 \quad : \underline{\mu}^{G2}, \overline{\mu}^{G2} \tag{1e}$$

$$p^{D1} + p^{D2} - p^{G1} - p^{G2} = 0 \quad : \lambda \tag{1f}$$

$$12 - \lambda - \underline{\mu}^{G1} + \overline{\mu}^{G1} = 0$$
$$0 \le p^{G1} \perp \underline{\mu}^{G1} \ge 0$$
$$0 \le (100 - p^{G1}) \perp \overline{\mu}^{G1} \ge 0$$

$$20 - \lambda - \underline{\mu}^{G2} + \overline{\mu}^{G2} = 0$$
$$0 \le p^{G2} \perp \underline{\mu}^{G2} \ge 0$$
$$0 \le (80 - p^{G2}) \perp \overline{\mu}^{G2} \ge 0$$

$$\lambda - 40 - \underline{\mu}^{D1} + \overline{\mu}^{D1} = 0$$
$$0 \le p^{D1} \perp \underline{\mu}^{D1} \ge 0$$
$$0 \le (100 - p^{D1}) \perp \overline{\mu}^{D1} \ge 0$$

$$\lambda - 35 - \underline{\mu}^{D2} + \overline{\mu}^{D2} = 0$$
$$0 \le p^{D2} \perp \underline{\mu}^{D2} \ge 0$$
$$0 \le (50 - p^{D2}) \perp \overline{\mu}^{D2} \ge 0$$

$$p^{D1} + p^{D2} - p^{G1} - p^{G2} = 0$$

Mixed complementarity problem (MCP)

Key takeaways

- The equilibrium and optimization forms of the market-clearing problem are equivalent because their corresponding KKT conditions are **identical**.

- Both forms lead to the "Nash equilibrium solution", meaning that no market player has the incentive to deviate from the market-clearing outcomes!

Key takeaways

- The equilibrium and optimization forms of the market-clearing problem are equivalent because their corresponding KKT conditions are **identical**.

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Question:

If, for any reason (such as externalities like tariffs, market power, etc.), the market-clearing optimization problem **no longer** has an equivalent equilibrium problem, can we still ensure that every market player will be satisfied with the clearing outcomes and will not unilaterally deviate?

Key takeaways

- The equilibrium and optimization forms of the market-clearing problem are equivalent because their corresponding KKT conditions are **identical**.

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Question:

If, for any reason (such as externalities like tariffs, market power, etc.), the market-clearing optimization problem **no longer** has an equivalent equilibrium problem, can we still ensure that every market player will be satisfied with the clearing outcomes and will not unilaterally deviate? No!

Form 1: Market clearing as an optimization problem

$$\underset{p_g^{\mathrm{G}},\ p_d^{\mathrm{D}},\ \theta_n}{\text{Maximize}} \quad \sum_d U_d\ p_d^{\mathrm{D}} - \sum_g C_g\ p_g^{\mathrm{G}}$$

subject to:

$$0 \le p_d^{\mathrm{D}} \le \overline{P}_d^{\mathrm{D}} \quad : \underline{\mu}_d^{\mathrm{D}}, \overline{\mu}_d^{\mathrm{D}} \quad \forall d$$

$$0 \le p_g^{\mathrm{G}} \le \overline{P}_g^{\mathrm{G}} \quad : \underline{\mu}_g^{\mathrm{G}}, \overline{\mu}_g^{\mathrm{G}} \quad \forall g$$

$$\sum_{d \in \Psi_n} p_d^{\mathrm{D}} + \sum_{m \in \Omega_n} B_{n,m}(\theta_n - \theta_m) - \sum_{g \in \Psi_n} p_g^{\mathrm{G}} = 0 \quad : \lambda_n \quad \forall n$$

$$-F_{n,m} \le B_{n,m}(\theta_n - \theta_m) \le F_{n,m} \quad : \underline{\eta}_{n,m}, \overline{\eta}_{n,m} \quad \forall n, \forall m \in \Omega_n$$

$$\theta_{(n=ref)} = 0 \quad : \gamma$$

Form 2: Market clearing as an equilibrium problem

Each generator:

$$\underset{p_g^{\mathrm{G}}}{\text{Maximize}} \quad p_g^{\mathrm{G}}(\lambda_{n:g\in\Psi_n} - C_g)$$

subject to:

$$0 \le p_g^{\mathrm{G}} \le \overline{P}_g^{\mathrm{G}} \quad : \underline{\mu}_g^{\mathrm{G}}, \overline{\mu}_g^{\mathrm{G}}$$

Each elastic demand:

$$\underset{p_d^{\mathrm{D}}}{\text{Maximize}} \quad p_d^{\mathrm{D}}(U_d - \lambda_{n:d\in\Psi_n})$$

subject to:

$$0 \le p_d^{\mathrm{D}} \le \overline{P}_d^{\mathrm{D}} \quad : \underline{\mu}_d^{\mathrm{D}}, \overline{\mu}_d^{\mathrm{D}}$$

Transmission owner as a spatial arbitrager (it buys power at a bus and sells it back at another one):

$$\underset{\theta_n}{\text{Maximize}} \quad \sum_{n,(m\in\Omega_n)} \lambda_n[B_{n,m}(\theta_m - \theta_n)]$$

subject to:

$$-F_{n,m} \le B_{n,m}(\theta_n - \theta_m) \le F_{n,m} \quad : \underline{\eta}_{n,m}, \overline{\eta}_{n,m} \quad \forall n, \forall m \in \Omega_n$$

Price-setter:

$$\underset{\lambda_n}{\text{Maximize}} \quad \sum_n \lambda_n \left(\sum_{d\in\Psi_n} p_d^{\mathrm{D}} + \sum_{m\in\Omega_n} B_{n,m}(\theta_n - \theta_m) - \sum_{g\in\Psi_n} p_g^{\mathrm{G}} \right)$$

Form 2: Market clearing as an equilibrium problem

Each generator:

$$\underset{p_g^{\mathrm{G}}}{\text{Maximize}} \quad \underbrace{p_g^{\mathrm{G}}(\lambda_{n:g\in\Psi_n} - C_g)}_{\text{Profit}}$$

subject to:

$$0 \le p_g^{\mathrm{G}} \le \overline{P}_g^{\mathrm{G}} \quad : \underline{\mu}_g^{\mathrm{G}}, \overline{\mu}_g^{\mathrm{G}}$$

Each elastic demand:

$$\underset{p_d^{\mathrm{D}}}{\text{Maximize}} \quad \underbrace{p_d^{\mathrm{D}}(U_d - \lambda_{n:d\in\Psi_n})}_{\text{Utility}}$$

subject to:

$$0 \le p_d^{\mathrm{D}} \le \overline{P}_d^{\mathrm{D}} \quad : \underline{\mu}_d^{\mathrm{D}}, \overline{\mu}_d^{\mathrm{D}}$$

Transmission owner as a spatial arbitrager (it buys power at a bus and sells it back at another one):

$$\underset{\theta_n}{\text{Maximize}} \quad \underbrace{\sum_{n,(m\in\Omega_n)} \lambda_n[B_{n,m}(\theta_m - \theta_n)]}_{\text{Congestion rent}}$$

subject to:

$$-F_{n,m} \le B_{n,m}(\theta_n - \theta_m) \le F_{n,m} \quad : \underline{\eta}_{n,m}, \overline{\eta}_{n,m} \quad \forall n, \forall m \in \Omega_n$$

Price-setter:

$$\underset{\lambda_n}{\text{Maximize}} \quad \sum_n \lambda_n \left(\sum_{d\in\Psi_n} p_d^{\mathrm{D}} + \sum_{m\in\Omega_n} B_{n,m}(\theta_n - \theta_m) - \sum_{g\in\Psi_n} p_g^{\mathrm{G}} \right)$$

A closer look at the formulation of congestion rent

Recall from the previous lecture that the transmission owner serves as a **spatial arbitrager** by buying power at one bus and selling it at another.

$$\underset{\theta_n}{\text{Maximize}} \quad \sum_{n,(m\in\Omega_n)} \lambda_n[B_{n,m}(\theta_m - \theta_n)]$$

subject to:

$$-F_{n,m} \leq B_{n,m}(\theta_n - \theta_m) \leq F_{n,m} \quad : \underline{\eta}_{n,m}, \overline{\eta}_{n,m} \quad \forall n, \forall m \in \Omega_n$$

Let's investigate whether this objective function (congestion rent) is correct!

A closer look at the producer's optimization problem

Each generator:

$$\underset{p_g^{\mathrm{G}}}{\text{Maximize}} \quad p_g^{\mathrm{G}}(\lambda_{n:g\in\Psi_n} - C_g)$$

subject to:

$$0 \leq p_g^{\mathrm{G}} \leq \overline{P}_g^{\mathrm{G}} \quad : \underline{\mu}_g^{\mathrm{G}}, \overline{\mu}_g^{\mathrm{G}}$$

Question: Can we prove the "cost recovery" of producers?

Cost Recovery: This concept implies that every market player achieves non-negative profits, meaning that market payments are sufficient to cover their costs. It is considered as a **desirable market property**.

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Question: Can we prove the "cost recovery" of producers?

Cost Recovery: This concept implies that every market player achieves non-negative profits, meaning that market payments are sufficient to cover their costs. It is considered as a **desirable market property**.

Question: Think of additional market players, like a storage owner. Can we extend the equilibrium problem to incorporate the profit-maximization problem of the storage owner?

Form 2: Market clearing as an equilibrium problem

Each generator:

$$\underset{p_g^{\mathrm{G}}}{\text{Maximize}} \quad p_g^{\mathrm{G}}(\lambda_{n:g\in\Psi_n} - C_g)$$

subject to:

$$0 \le p_g^{\mathrm{G}} \le \overline{P}_g^{\mathrm{G}} \quad : \underline{\mu}_g^{\mathrm{G}}, \overline{\mu}_g^{\mathrm{G}}$$

Each elastic demand:

$$\underset{p_d^{\mathrm{D}}}{\text{Maximize}} \quad p_d^{\mathrm{D}}(U_d - \lambda_{n:d\in\Psi_n})$$

subject to:

$$0 \le p_d^{\mathrm{D}} \le \overline{P}_d^{\mathrm{D}} \quad : \underline{\mu}_d^{\mathrm{D}}, \overline{\mu}_d^{\mathrm{D}}$$

Transmission owner as a spatial arbitrager (it buys power at a bus and sells it back at another one):

$$\underset{\theta_n}{\text{Maximize}} \quad \sum_{n,(m\in\Omega_n)} \lambda_n[B_{n,m}(\theta_m - \theta_n)]$$

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$$-F_{n,m} \le B_{n,m}(\theta_n - \theta_m) \le F_{n,m} \quad : \underline{\eta}_{n,m}, \overline{\eta}_{n,m} \quad \forall n, \forall m \in \Omega_n$$

Price-setter:

$$\underset{\lambda_n}{\text{Maximize}} \quad \sum_n \lambda_n \left(\sum_{d\in\Psi_n} p_d^{\mathrm{D}} + \sum_{m\in\Omega_n} B_{n,m}(\theta_n - \theta_m) - \sum_{g\in\Psi_n} p_g^{\mathrm{G}} \right)$$

Form 3: Market clearing as an MCP

$$-U_d + \lambda_{n:d\in\Psi_n} - \underline{\mu}_d^{\mathrm{D}} + \overline{\mu}_d^{\mathrm{D}} = 0 \quad \forall d$$

$$C_g - \lambda_{n:g\in\Psi_n} - \underline{\mu}_g^{\mathrm{G}} + \overline{\mu}_g^{\mathrm{G}} = 0 \quad \forall g$$

$$\sum_{m\in\Omega_n} B_{n,m}(\lambda_n - \lambda_m + \overline{\eta}_{n,m} - \overline{\eta}_{m,n} - \underline{\eta}_{n,m} + \underline{\eta}_{m,n}) + (\gamma)_{n=ref} = 0 \quad \forall n$$

$$\sum_{d\in\Psi_n} p_d^{\mathrm{D}} + \sum_{m\in\Omega_n} B_{n,m}(\theta_n - \theta_m) - \sum_{g\in\Psi_n} p_g^{\mathrm{G}} = 0 \quad \forall n$$

$$\theta_{(n=ref)} = 0$$

$$0 \le p_g^{\mathrm{G}} \perp \underline{\mu}_g^{\mathrm{G}} \ge 0 \quad \forall g$$

$$0 \le [\overline{P}_g^{\mathrm{G}} - p_g^{\mathrm{G}}] \perp \overline{\mu}_g^{\mathrm{G}} \ge 0 \quad \forall g$$

$$0 \le p_d^{\mathrm{D}} \perp \underline{\mu}_d^{\mathrm{D}} \ge 0 \quad \forall d$$

$$0 \le [\overline{P}_d^{\mathrm{D}} - p_d^{\mathrm{D}}] \perp \overline{\mu}_d^{\mathrm{D}} \ge 0 \quad \forall d$$

$$0 \le [F_{n,m} + B_{n,m}(\theta_n - \theta_m)] \perp \underline{\eta}_{n,m} \ge 0 \quad \forall n, \forall m \in \Omega_n$$

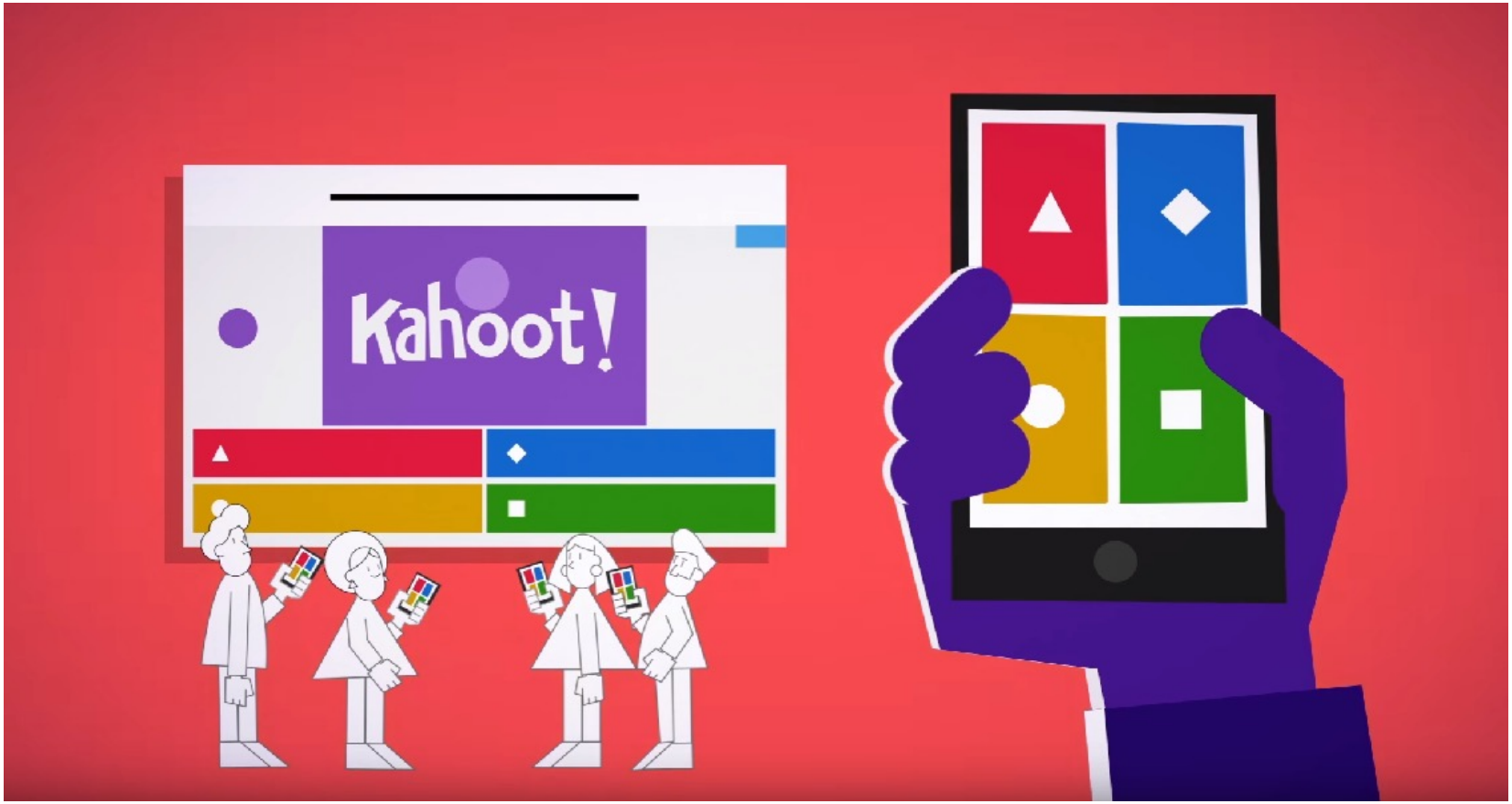
$$0 \le [F_{n,m} - B_{n,m}(\theta_n - \theta_m)] \perp \overline{\eta}_{n,m} \ge 0 \quad \forall n, \forall m \in \Omega_n$$

Conclusion

All three forms of the market-clearing problem, i.e.,

- optimization
- equilibrium
- MCP

are equivalent!


Kahoot!

Thanks for your attention!

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